One-Sample Mean Inference

Estimating a population average

Download the section 13 .Rmd handout to `STAT240/lecture/sect13-single-mean`.

Material in this section is covered by Chapter 13 on the notes website.

Download the file `cat_breeds_clean.csv` to `STAT240/data`.

`cat_breeds_clean.csv` contains characteristics of 1071 domestic cats (including daily hours of sleep).

Possible questions of interest:

- What is the average daily hours of sleep μ?
- Is there a difference between μ_{fixed} and μ_{intact}?

Inference for μ:

- Point estimate: $\bar{X}$
- Standard error:

$$\frac{S}{\sqrt{n}}$$

- Model: T distribution (a "wide normal")

Analysis is based on the CLT.

We assume the observations, X_i are i.i.d. with mean μ and sd σ.

$$X_i \sim D(\mu, \sigma)$$

Let's focus on adult male fixed ragdoll cats.

The point estimate for μ is the sample mean

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

and we observed $\bar{x} = 16.02$.

Let's look at a simulation for $\bar{X}$.

The CLT gives

$$\bar{X} \dot{\sim} N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

for large enough n, which will be the basis of our CI and H-test.

A CI, in general, is

$$\text{point estimate} \ \pm \ \text{critical value} \ \times \ \text{standard error}$$

- For μ, the point estimate is $\bar{x}$
- The standard error is $\frac{\sigma}{\sqrt{n}}$

We should be able to use a Z critical value:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Problem: we don't have σ. The quantity

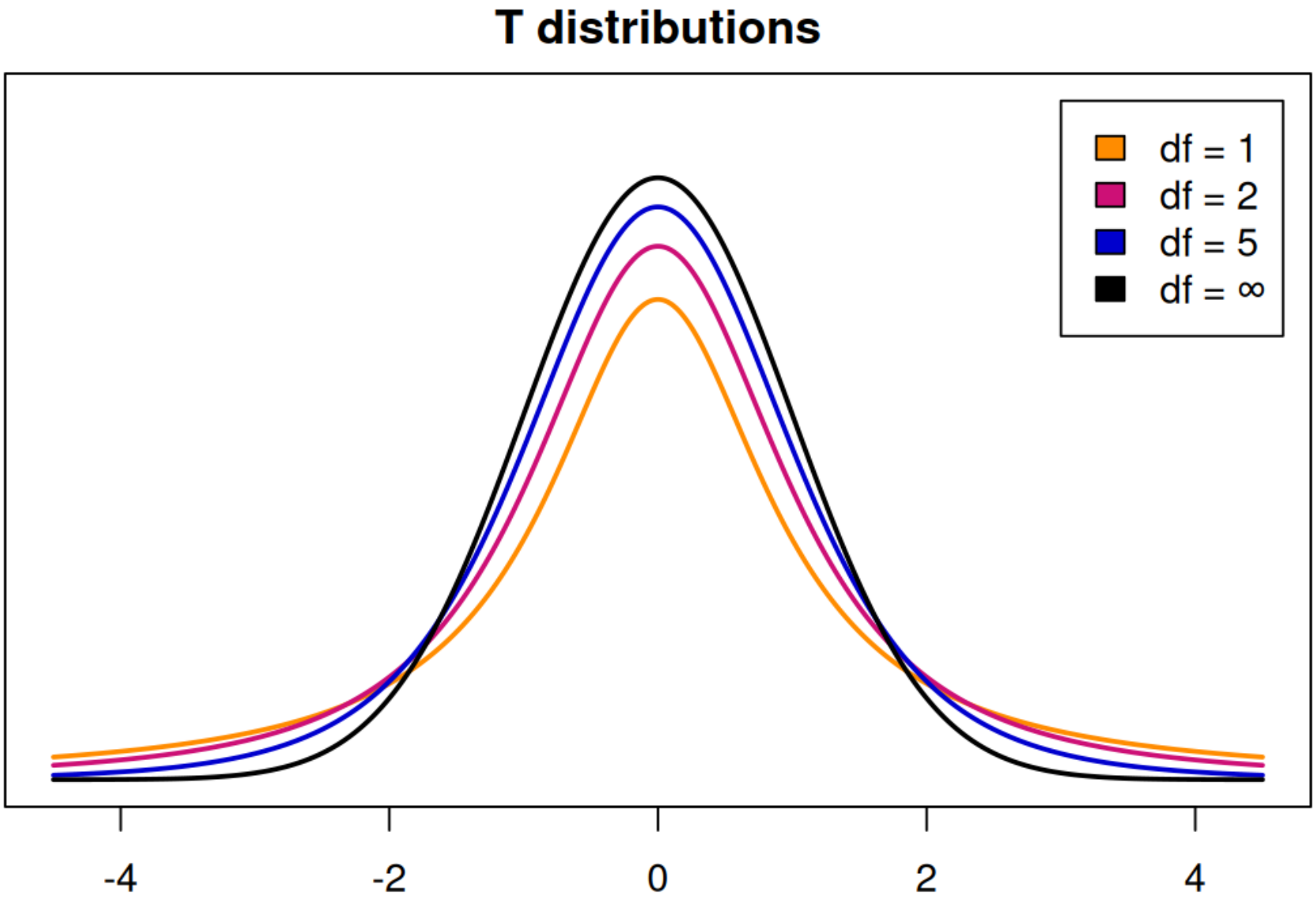
$$\frac{\bar{X} - \mu}{S/\sqrt{n}}$$

is not *quite* normal.

The sampling distribution for $\bar{X}$ when σ is unknown uses the **Student's T distribution**.

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim T$$

This is similar to $N(0, 1)$, but wider.

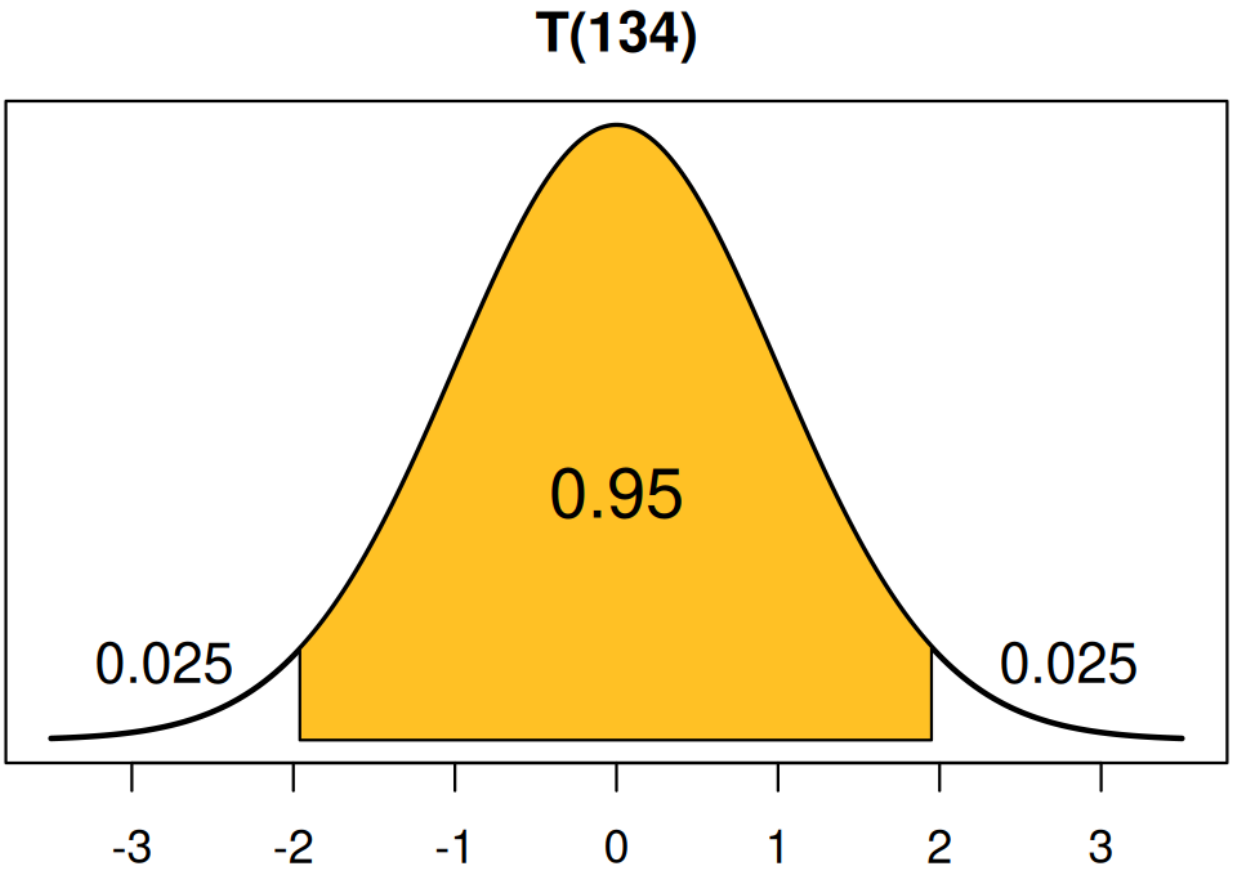

T distributions
df = 1
df = 2
df = 5
df = ∞
-4
-2
0
2
4

The T has heavier tails than $N(0, 1)$, controlled by degrees of freedom.

In single mean inference, $df = n - 1$.

Find critical values (quantiles) with `qt()`.

For 95% confidence:



T CI for μ:

$$\bar{X} \pm t_{\alpha/2,n-1} \cdot \frac{S}{\sqrt{n}}$$

We are 95% confident that the true average daily hours of sleep for this group of cats is within (15.63, 16.41).

Check with `t.test()`.

Is the true average daily sleep for this group equal to 17 hours?

$$H_0 : \mu = 17 \quad \text{versus} \quad H_A : \mu \neq 17$$

We gather evidence against H_0 with $\bar{x} = 16.02$.

Let's use $\alpha = 0.05$.

If $\mu = 17$, then the quantity

$$\frac{\bar{X} - 17}{\sigma/\sqrt{n}}$$

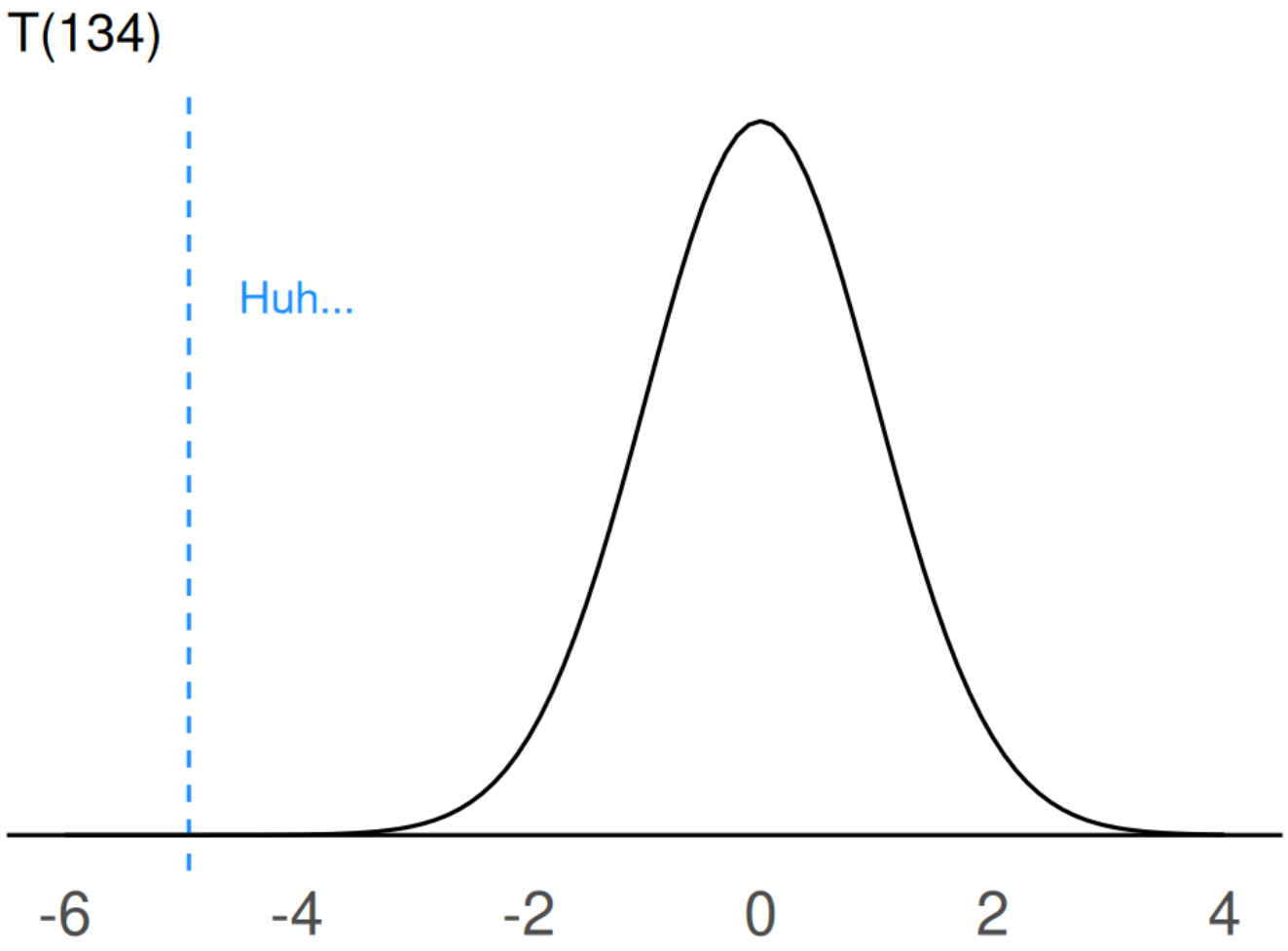
should be $N(0, 1)$, implying a Z test.

But, we don't know σ.

Instead, use

$$\frac{\bar{X} - 17}{S/\sqrt{n}} \sim T_{n-1}$$

Our observed test statistic is -4.93. Is this consistent with our null T distribution?


T(134)
Huh...
-6
-4
-2
0
2
4

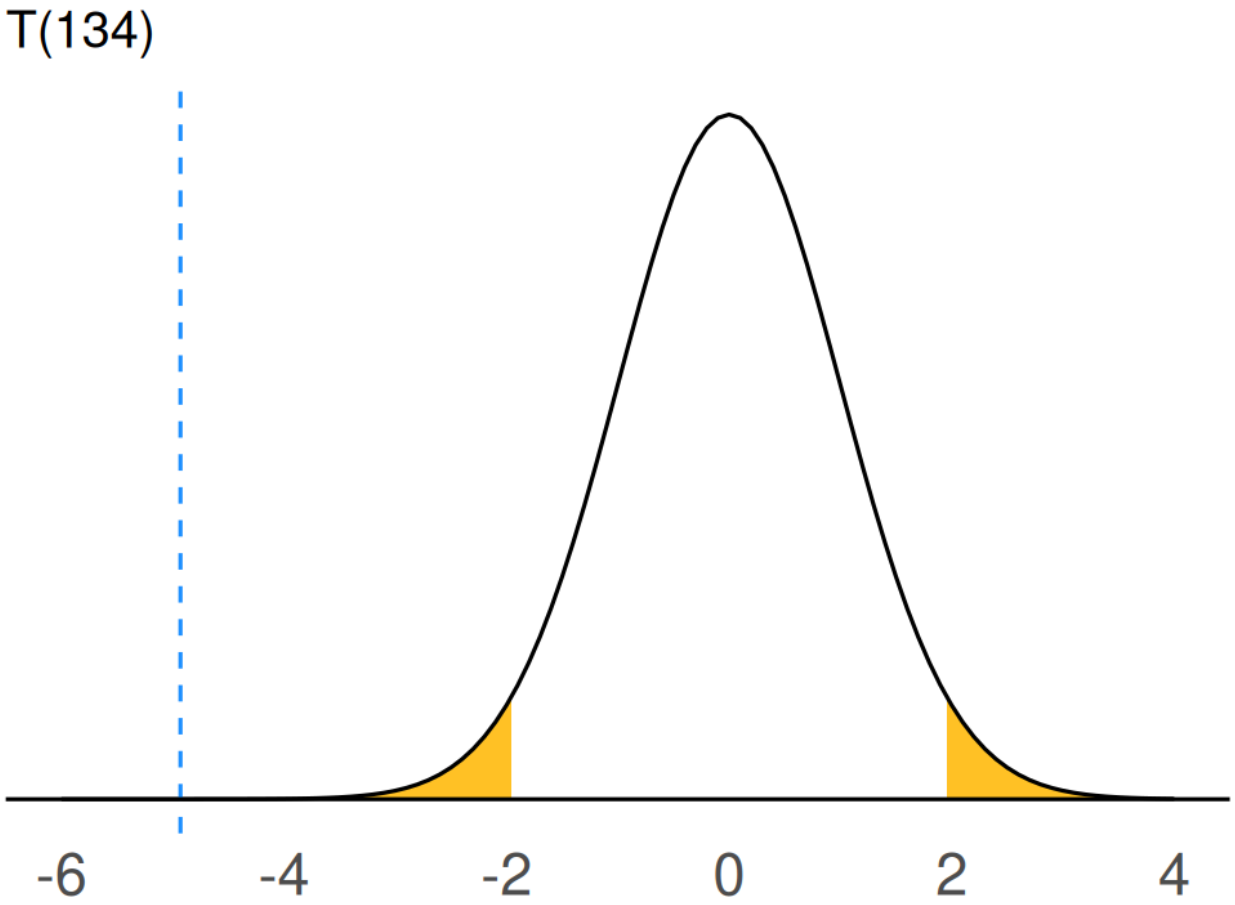
Our p-value

$$P(T_{n-1} \leq -4.93) \quad + \quad P(T_{n-1} \geq 4.93)$$

is very small, so we reject H_0.

We can check this result in `t.test()`.

A rejection region also rejects H_0.



Recall that our 95% CI for μ did not contain 17: (15.63, 16.41).

Similarly, we rejected the null value 17 at the 5% level. In fact, the methods are equivalent.

The following 2 statements are equivalent for T CIs and hypothesis tests for μ:

- A $100(1-\alpha)\%$ CI does not contain μ_0.
- A test for $H_A : \mu \neq \mu_0$ at level α rejects H_0.

Conversely, if μ_0 is inside of the $100(1-\alpha)\%$ CI, we would fail to reject H_0 in a two-sided test.

Consider adult (fixed) female ragdoll cats. Do we have evidence that their average daily sleep time is *less* than 17 hours?

- Write one-sided hypotheses about μ
- Re-calculate summary statistics
- Perform the T test with $\alpha = 0.02$.
- Check with `t.test()`

In general, perform a T test on independent draws from the same population. Use hypotheses

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_A : \mu \neq \mu_0$$

and test statistic

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

and null distribution T_{n-1}.

P-value calculation:

- If the alternative is $H_A : \mu < \mu_0$, the p-value is the area *below* the test statistic.
- If the alternative is $H_A : \mu > \mu_0$, the p-value is the area *above* the test statistic.
- If the alternative is $H_A : \mu \neq \mu_0$, the p-value is $2\times$ the area *outside* of the test statistic.